

Jacob Zheng

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EDUCATION

University of Waterloo

Bachelor of Mathematics, Honours, Co-op

Waterloo, ON

Expected Graduation Apr 2030

- **President's Scholarship** - awarded to incoming students with an admission average above 90%
- **Clubs:** WAT.AI , UW Orbit

TECHNICAL SKILLS

Languages: TypeScript, Python, JavaScript, C, SQL

Frameworks / Libraries: Node.js, React, Next.js, TailwindCSS, FastAPI, NumPy, PyTorch, PostgreSQL

Developer Tools: Git, GitHub, VS Code, Linux, Docker

PROJECTS

Multimodal AI Teaching Assistant ↗

Hack Canada

- Delivered live spoken tutoring feedback at **under 75ms latency**, as measured by end-to-end API response benchmarks, by leading frontend development in **React** and **Next.js** and integrating the **Eleven-Labs Conversational AI** and **Gemini APIs** to detect and analyze student mistakes on a **TLDraw whiteboard** in real time.
- Achieved **90% accuracy** in detecting student disengagement, validated against labeled test sessions, by implementing a real-time facial engagement monitor using **MediaPipe Face Mesh** directly in the browser to process **468 facial landmarks** per frame.
- Eliminated service desyncs during live sessions, confirmed by zero-dropout testing across extended runs, by connecting three external APIs (**ElevenLabs Scribe**, **Gemini**, **TLDraw**) through WebSocket reconnection logic and in-memory state management within a stateless **FastAPI** session architecture.

Cross-Cultural Music Trends Analyzer

- Built a **data pipeline** to collect and aggregate top chart data across **50+ countries** via the Spotify API, capturing millions of data points on regional listening patterns.
- Applied **clustering algorithms** to group nations by musical similarity, surfacing cross-cultural listening trends across audio features like BPM, key, and energy.
- Designed an **interactive world map dashboard** visualizing country-level music taste.
- Analyzed correlations between music taste clusters and **geographic, linguistic, and cultural factors** to uncover what drives regional listening habits.

EXPERIENCE

Ski Instructor

Dec 2022 – Jan 2025

Camp Fortune

Chelsea, QC

- Instructed **30+ students per day**, adapting to different skill levels while maintaining **CSIA Level 1**.
- Nominated **Best Instructor of the Year** twice by management.

Court Attendant

Sept 2022 – Aug 2025

Barrhaven Tennis Club

Ottawa, ON

- Promoted from member to Court Attendant; took on a leadership role recruiting and onboarding new staff.
- Assisted **50+ members per shift** across bookings, events, and tournaments at a **250-member** facility.